

Depth and camera geometry

A pixel identifies a viewing direction. Metric geometry needs camera calibration, depth, coordinate conventions, and valid alignment.

15–20 MINUTES

Prerequisite: image coordinates

DEPTH REASONING

1 • Pixel

A location on the image plane.

2 • Ray

A direction through the camera.

3 • Depth

A distance convention along that ray.

4 • 3-D point

Back-project with intrinsics.

Why one image loses scale

A small nearby object and a large distant object can create the same image size. Perspective projection maps a 3-D point to a 2-D pixel, discarding one dimension. A bounding box therefore supplies an image location, not physical size or distance by itself.

Central mental model

Pixel + calibrated camera model gives a ray. Pixel + valid depth + calibration gives a 3-D point in a named coordinate frame.

Camera intrinsics

The pinhole model uses focal lengths f_x , f_y and principal point c_x , c_y . For a pixel (u, v) with z-depth Z , a common back-projection is:

$$X = (u - c_x) \times Z / f_x$$

$$Y = (v - c_y) \times Z / f_y$$

$$Z = \text{depth along the optical axis}$$

These values are meaningful only when units and frames are known. Calibration parameters belong to a particular camera stream and resolution. Resizing or cropping images requires corresponding coordinate handling.

Depth conventions and invalid data

ROS REP 118 defines canonical depth as the distance along the camera's optical Z axis, in meters for floating-point images. A common raw `uint16` representation uses millimeters and zero for invalid depth. Other sensors or APIs may use different conventions, so inspect the message encoding and documentation.

RGB and depth pixels must also refer to the same rays. “Same width and height” does not prove registration. If depth is aligned to another camera, using an RGB box directly can sample the wrong physical region.

Browser exercise: reason about z-depth

A floor point behind a cube is at 1200 mm along optical Z. The visible cube top is closer to the camera at 1050 mm.

Which comparison selects the closer cube top?

Physical mounting can make the scene more complicated, but the sign must follow the declared depth convention and be verified on registered sensor data.

From depth samples to geometry evidence

A box contains foreground, background, edges, invalid values, and possible occluders. Robust statistics such as medians or percentiles can be more useful than one pixel. A geometry check should also define behavior when too few valid samples exist. Keeping the detection favors recall; rejecting it favors strict validation.

Applied to our cubes

REPOSITORY OBSERVATION The node synchronizes RGB and depth messages and uses camera intrinsics to estimate metric extents from a detected box.

CONFIRMED DEFECT The current geometry selection uses `depth > floor + threshold`. For ordinary z-depth, that selects farther points and must not be described as validated raised-surface reasoning.

UNRESOLVED ASSUMPTION RGB/depth registration and the filter's physical interpretation need validation against actual hardware evidence.

Project anchors: geometry filter and RGB/depth callback.

Check your understanding

What is needed beyond pixel coordinates for a metric 3-D point?

- ☐ A larger confidence threshold
- ☐ A second class prediction
- ☐ Depth and camera calibration

Explain it back

Explain the difference between a pixel, a viewing ray, z-depth, and a 3-D point. Then audit the sign of the current “raised” comparison.

Primary sources

- Szeliski — image formation, calibration, and 3-D reconstruction.
- ROS REP 118 — depth conventions and encodings.